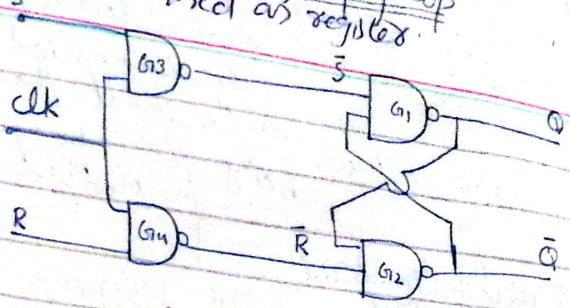
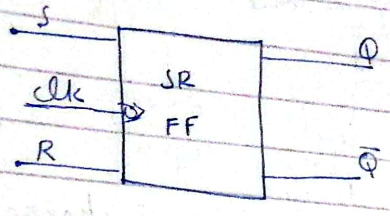
SR flip flop

FFs are used as register.





$S' = \overline{S \cdot CLK} \Rightarrow \bar{S} + \overline{clk}$ —— (i)

$R' = \overline{R \cdot CLK} = \bar{R} + \overline{clk}$ —— (ii)

case I:

when clk = 0

$S' = \bar{S} + 1 = 1$

$R' = \bar{R} + 1 = 1$

Case II

clk = 1

$S' = \bar{S} + 0 = \bar{S}$

$R' = \bar{R} + 0 = \bar{R}$

Truth table

i) $S = 0, R = 1$; $\bar{S} = 1, \bar{R} = 0$

ii) $S = 1, R = 0$; $\bar{S} = 0, \bar{R} = 1$

CLK	S	R	Q	$\bar{Q}$
0	X	X	Prev state	Prev state
1	0	0	No change	No change
1	0	1	0	1
1	1	0	1	0
1	1	1	Invalid	Invalid

Note: we ... SR latch NAND g... and t...

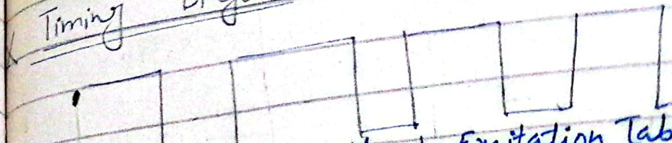
Final truth table for SR FF without

CLK	S	R	Q_{n+1}
0	X	X	Q_n
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	Invalid

Truth table

Truth table is a table b/w i/p & o/p

Timing Diagram:



Characteristics Table

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

Excitation Table

Q_n	Q_{n+1}	S_n	R_n
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Excitation table

– table b/w current state and next state w.r.t i/p.

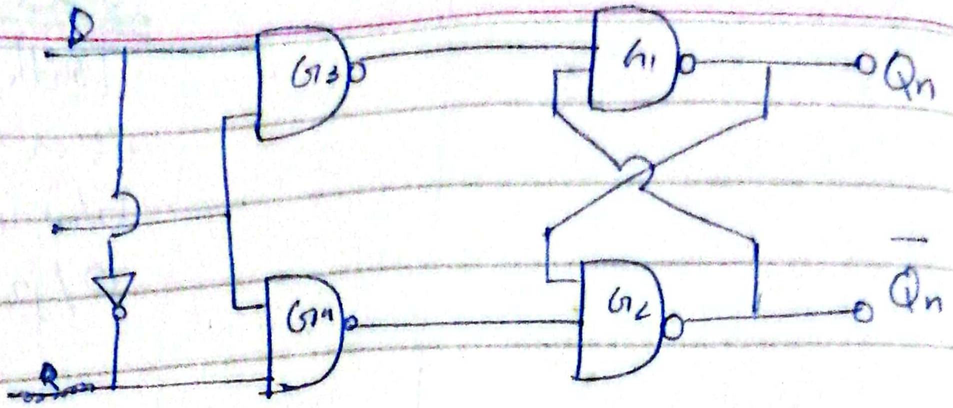
R_n \ $Q_n S_n$	00	01	11	10
0	0	1	1	1
1	0	X	X	0

$Q_{n+1} = S_n + Q_n R_n$

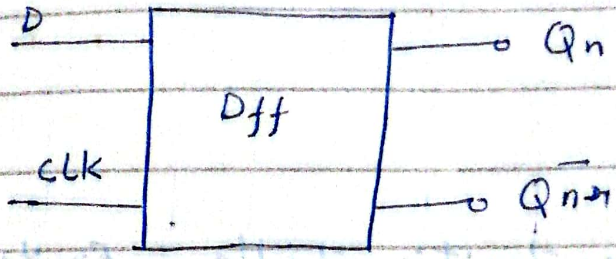
Characteristic Table

⇒ is a table b/w current state and input w.r.t next state.

D flip flop - is data storage flip flop.



Block Diagram

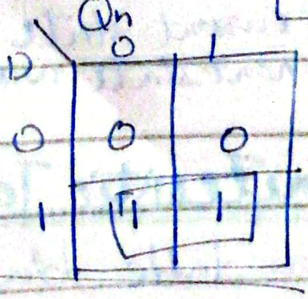


CLK	D	Q_{n+1}
0	X	Qn
1	0	0
1	1	1

characteristic Table

Qn	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

$\boxed{Q_{n+1} = D}$ → characteristic equat



D \ Qn	0	1
0	0	0
1	1	1

Excitation table

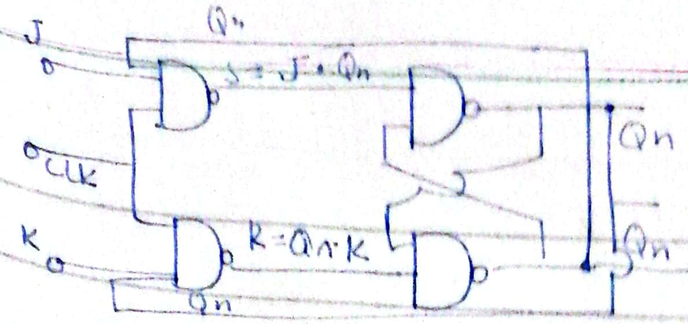
→ used for Previous & next state.

Present state	Next state	D
Qn	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

$\boxed{D = Q_{n+1}}$

excitation table

JK flip flop



→ if J and K are both low then no change occurs.

→ if J and K are both high at the clock edge then the output will be toggle from one state to another.

- JK flip flops can function as set or Reset flip fl.

SR flipflop TT

CLK	S	R	Q_{n+1}
0	x	x	Q_n
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	Invalid

Truth table

J	K	Q_n	$\overline{Q_n}$	S	R	Q_{n+1}	
0	0	0	1	0	0	~~Qn~~ Q_n	Q_n
0	0	1	0	0	0	Q_n	
0	1	0	1	0	0	0	0
0	1	1	0	0	1	0	
1	0	0	1	1	0	1	1
1	0	1	0	0	0	1	
1	1	0	1	1	0	1	$\overline{Q_n}$
1	1	1	0	0	1	0	

Race around condition:

change

Final truth table for JK flip flop

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\bar{Q}_n$

characteristic table

Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

K \ $Q_n J$	00	01	11	10
0	0	1	1	1
1	0	1	0	0

$$\boxed{Q_{n+1} = \bar{Q}_n J + Q_n \bar{K}}$$

↓

characteristic equation

Excitation table

Present state Q_n	Next state/o/p Q_{n+1}	I/p J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

$$\boxed{K = \overline{Q_{n+1}}}$$

J

Q_{n+1} \ Q_n	0	1
0	0	X
1	1	X

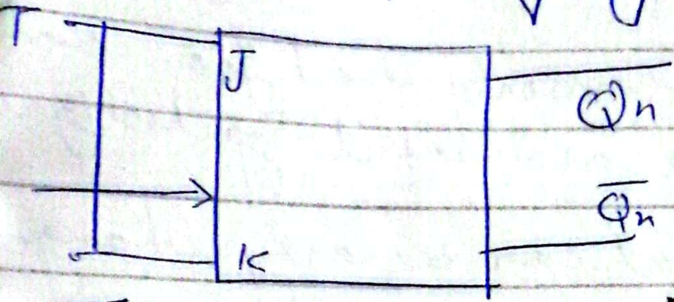
$$\boxed{J = Q_{n+1}}$$

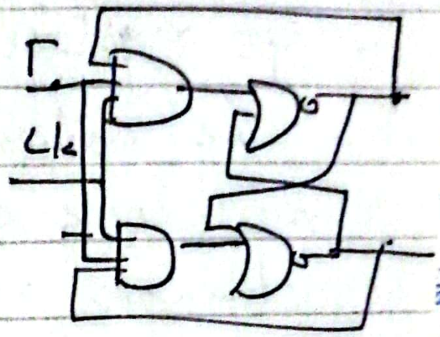
K

Q_{n+1} \ Q_n	0	1
0	X	1
1	X	0

T flip flop

- T stands for toggling





J=K

Truth table

QtK	T	Q_{n+1}
0	X	Q_n
1	0	Q_n
1	0	$\bar{Q}_n$

means J=0, K=0

Characteristics table

Q_n	T	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

→ same

$Q_{n+1} = Q_n \oplus T$

Excitation table

J	K
0	X
1	X
X	0
X	1

for J

T \ Q_n	0	1
0	0	X
1	1	X

J = T

for K

T \ Q_n	0	1
0	X	0
1	X	1

K = T

~~Population~~

D-flip flop: D-flip flop is better alternative that is very popular with digital electronics

- They are commonly used for counters, shift registers and input synchronization.
- The output can be only changed at the clock edge, and if the output changes at other times the output will be unaffected.

T-flip flop: is like a J.K flip flop. These are basically single input version of JK flipflops. This modified from JK flip flop is obtained by connecting both inputs J and K together. It has only one input along with clock input.

- These flip-flops called T-flip flops because of their ability to complement its state. i.e Toggle hence the name Toggle flip flops

JK flip flops

- if J & K data input are different (high & low) then the output Q takes the value of J at the next clock edge.

Flip flop Operating Characteristics :-

The performance of the flip flop is specified by several operating characteristics mentioned in the data sheet of the flip flops.

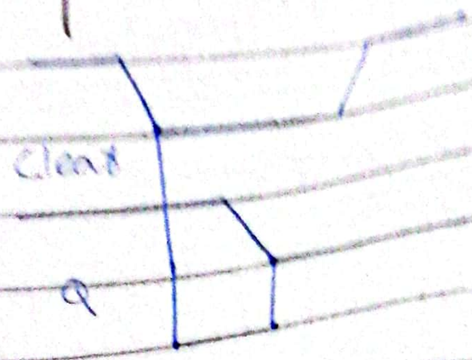
The important operating characteristics are

- → Propagation delay
- → Set-up time
- → Hold time
- → Maximum clock frequency
- → Pulse with
- → Power Dissipation.

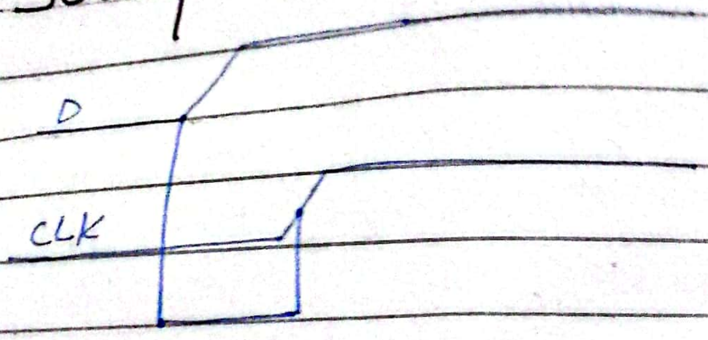
Propagation Delay:-

→ The propagation delay time is the interval of time when the input is applied and the output changes.

Set-up time:-



Set-up time:-

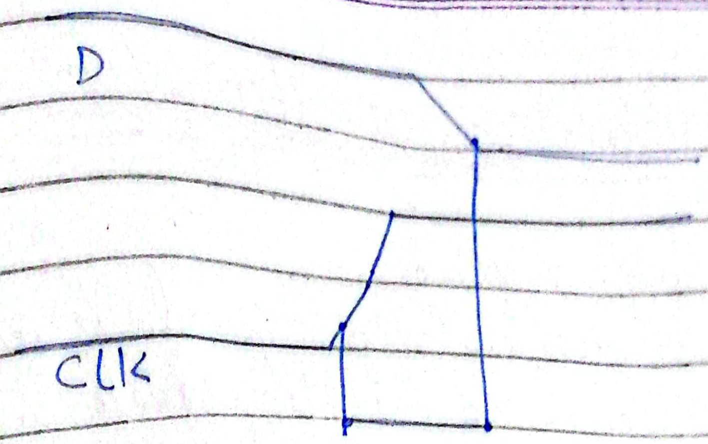


→ The minimum number of time that stable input before the occurance of clock transition is called setup time

Hold time:-

The minimum time for which the input signal has to be maintained at the input is the hold time of the flip flop.

CamScanner

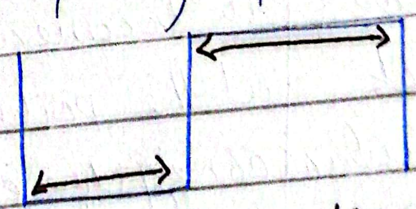


Maximum clock frequency:-

The maximum clock frequency f_{max} is the highest rate at which the flip flop operates reliably.

Pulse width:-

Signal has to be of a specified duration for the correct operation of flip flop



Power Dissipation:-

How much energy is used by a circuit to perform operation.

$$P = V * I$$

Application of flip-flops

-> Shift registers
- Parallel to serial conversion
- Serial to parallel conversion
- Data storage → serial / → parallel

-> Counters
- Count the number of pulses
- Frequency Division

Differentiate Level triggering & edge triggering?

Level triggering	Edge triggering
- Allows a circuit to become active when the clock pulse on a particular level.	- Allows a circuit to become active at the positive edge or negative edge of the clock signal.
- An event occurs at rising edge or falling edge.	- An event occurs during the high voltage or low volt
- Latches are level triggered	- flip flops are tri

2. Write in a ...
3. You are not allowed to bring ...
4. Use your own stationery items.

Examiner Remarks

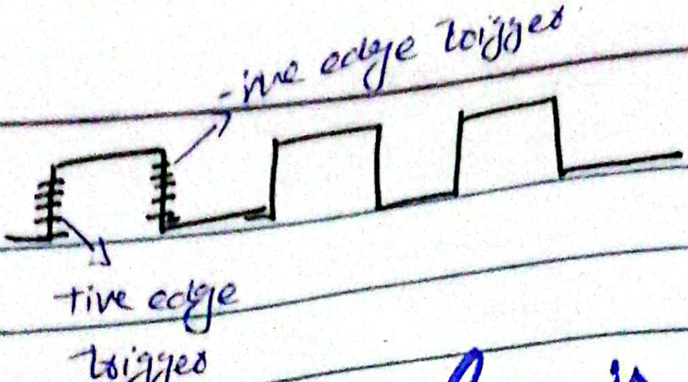
Differentiate falling and rising edge;

Falling edge:-

→ The edge that changes the voltage from high 90% level to the low level is called falling edge (negative edge)

Rising edge:-

The edge that changes the voltage from low to high level is called rising edge (+ive edge).

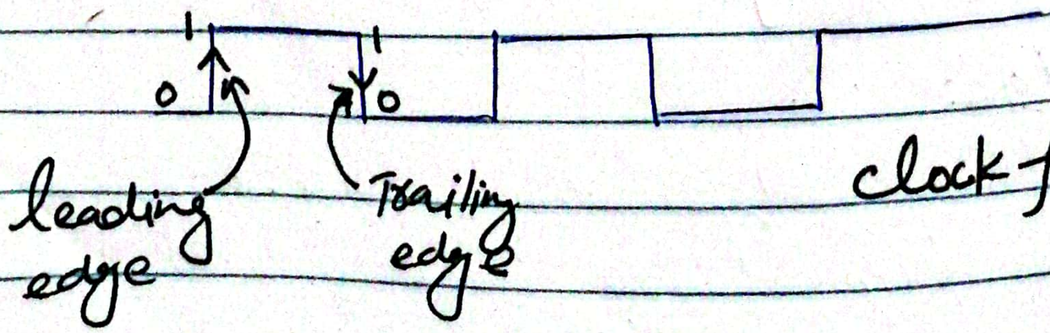


why D flip flop is called synchronous?

The D input of the D flip flop is synchronous because data on the input are transferred to the flip-flop o/p only on the triggering edge of the clock pulse.

Define clock Signal:

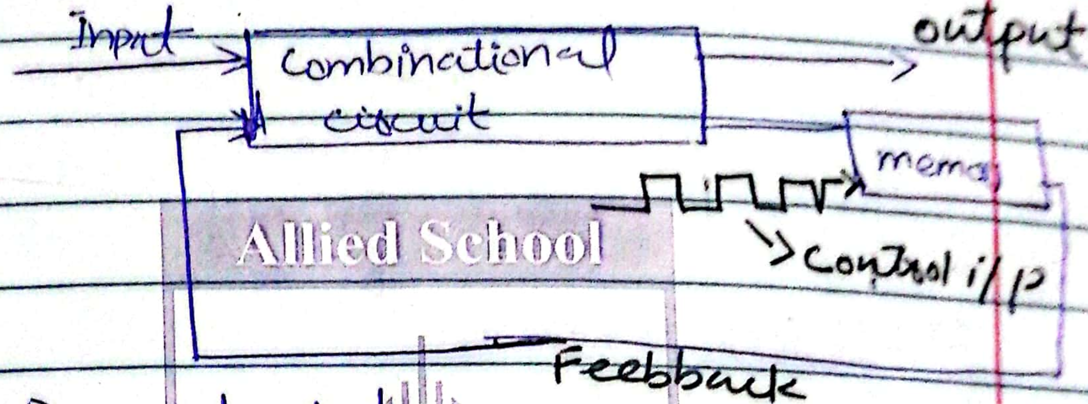
- Clock signal is a timing signal.
  (- Also called clock pulse
- It is used to provide the sequence of the circuit.
- It work as control signal for sequential circuit.



clock freq = $\frac{1}{T}$

Define term duty cycle.

= ratio of time for which the signal is high / total time



Qn = Present state

Qn+1 = State after clock pulse.

Define trigger pulse.

The change in output of a flip flop can be done by bringing a small change in the input signal. This small change can be done by [bringing a small change] clock pulse. This clock pulse is called trigger pulse.

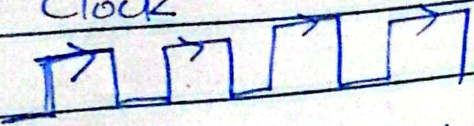
Triggering → when flip flop is on and give output

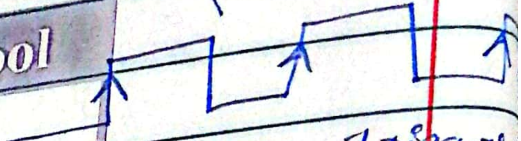
Level

Edge

High level triggering

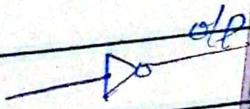
Clock

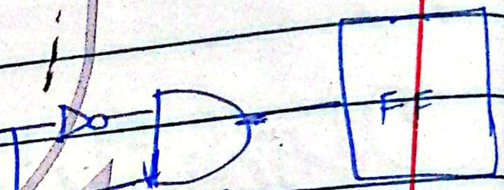




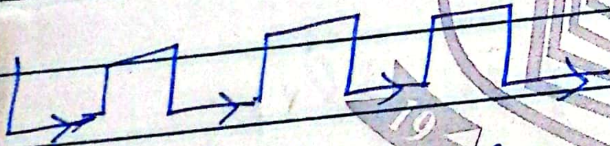
+ive edge triggering

clck

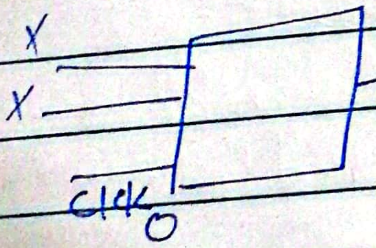




+ive edge triggering

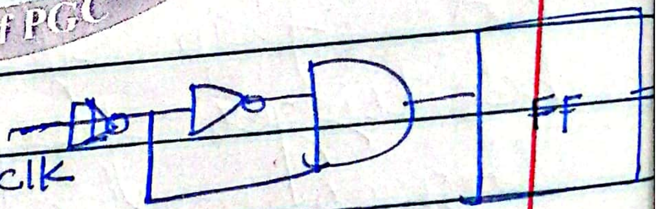


Low level triggering

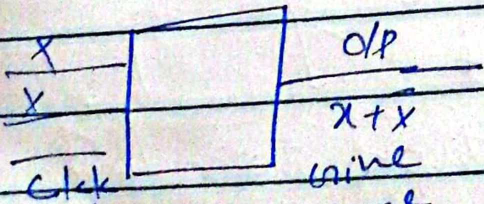


No change

clk

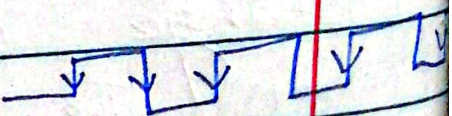


-ive edge triggering



x+x
wire
some
value

-High level triggering



Then output will depend on input

Examiner's Remarks

→ when the clock changes from a low state to high state this is called the +ive-going transition or +ive edge triggered.

→ when the clock changes from a High state to low state, it is called negative going transition or negative edge triggered.

Define trigger flip flop.

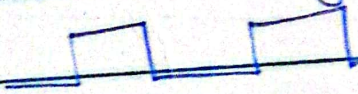
→ A flip flop is said to be trigger flip flop when a trigger pulse is applied to the input that bring the change in output.

Present state: The information stored in the memory elements at any given time defines the present state of the sequential circuit

Next state: The present state and external inputs determine the output and the next state of the sequential circuit.

Differentiate:-

Dif between flip flops & latches.

Flip-flops	Latches
- Synchronous	- Asynchronous
- Bistable device	- Bistable device
- faster	- slower
0 or 1	1 0 or 1
- clock signal 	no clock
- require less power as it does not have clock signal	- require more power
3- Edge triggered, Pulse - triggered	- level triggered - gated latches - Enable
-4 Particular time I/P → O/P	continuously input change then O/P will also change
- it is structured based on logic gates block	- it is structured based on blocks of latch and logic
- Designing is less complex	- more complex.

Digital Logic Design:-